



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0905 JAVA PROGRAMMING – C03-AT1

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Secure Exit

SIMATS Online Programming Lab
JAVA PROGRAMMING

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Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     static int balance;
4     static void withdraw(int amount)throws InsufficientBalanceException
5     {
6         if(amount>balance){
7             throw new InsufficientBalanceException("Withdrawal amount exceeds available balance.");
8         }
9         balance-=amount;
10    }
11    static void validateBalance(int amount) throws InsufficientBalanceException
12    {
13        withdraw(amount);
14    }
15    static void processTransaction(int amount) throws InsufficientBalanceException
16    {
17        validateBalance(amount);
18    }
19    public static void main(String[] args){
20        Scanner sc=new Scanner(System.in);
21        balance=sc.nextInt();
22        int withdrawAmount=sc.nextInt();
23        try{
24            processTransaction(withdrawAmount);
25            System.out.println("Transaction successful");
26        }
27        catch(InsufficientBalanceException e){
28            System.out.println("Exception propagated to main().");
29            System.out.println("InsufficientBalanceException: "+e.getMessage());
30        }
31    }
32 }
```

Input

5000
7000

Output

```
Exception propagated to main().
InsufficientBalanceException:
Withdrawal amount exceeds
available balance.
```

Visible Test Cases

Test Case 1	✗
Test Case 2	✗
Test Case 3	✗

0.00 / 10 marks

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WeakCrosswordException.
10 marks

Q4
Email Validation
10 marks

Q5
Banking Transaction Using
Exception Propagation
10 marks

5/5 answered

Exception: Invalid email address.

Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         String email=sc.nextLine();
6         try{
7             if(!email.contains("@")){
8                 throw new InvalidEmailException("Invalid email address");
9             }
10            System.out.println("Valid email address");
11        }
12        catch(InvalidEmailException e){
13            System.out.println("Exception: "+e.getMessage());
14        }
15    }
16 }
17 class InvalidEmailException extends Exception{
18     public InvalidEmailException(String message){
19         super(message);
20     }
21 }
```

Input

student@gmail.com

Output

```
Exception: Invalid email address.
```

Visible Test Cases

Test Case 1	✓
Test Case 2	✓
Test Case 3	✓

10.00 / 10 marks

Submitted

Q4
Email Validation
10 marksQ5
Banking Transaction Using
Exception Propagation
10 marks

5/5 answered

Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         String password=sc.nextLine();
6         try{
7             if(password.length()<8){
8                 throw new WeakPasswordException("Password must contain at least 8 characters.");
9             }
10            System.out.println("Password is strong");
11        }
12        catch(WeakPasswordException e){
13            System.out.println("Exception: "+e.getMessage());
14        }
15    }
16 }
17 class WeakPasswordException extends Exception{
18     public WeakPasswordException(String message){
19         super(message);
20     }
21 }
```

Submitted

Input

java123

Output

Exception: Password must contain at least 8 characters.

Visible Test Cases

Test Case 1	✓
Test Case 2	✓
Test Case 3	✓

10.00 / 10 marks

Q2
IllegalArgumentException
10 marksQ3
user-defined exception named
WeakPasswordException.
10 marksQ4
Email Validation
10 marksQ5
Banking Transaction Using
Exception Propagation
10 marks

5/5 answered

Write a Java program to accept marks (0–100) for a student. If the entered mark is less than 0 or greater than 100, throw an `IllegalArgumentException` and display an appropriate message.

Sample Input

120

Sample Output

Exception: Invalid marks. Marks should be between 0 and 100.

Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int marks=sc.nextInt();
6         try{
7             if (marks<0||marks>100){
8                 throw new IllegalArgumentException();
9             }
10            System.out.println("Valid marks:"+marks);
11        }
12        catch(IllegalArgumentException e){
13            System.out.println("Exception: Invalid marks. Marks should be between 0 and 100.");
14        }
15    }
16 }
```

Input

120

Output

Exception:Invalid marks. Marks should be between 0 and 100.

Visible Test Cases

Test Case 1	✓
Test Case 2	✓

Q3: Custom Exception named
WeakPasswordException.
10 marks

Q4
Email Validation
10 marks

Q5
Banking Transaction Using
Exception Propagation
10 marks

5/5 answered

Exception: Invalid marks. Marks should be between 0 and 100.

Your Code

```
1 import java.util.Scanner;  
2 public class Main{  
3     public static void main(String[] args){  
4         Scanner sc=new Scanner(System.in);  
5         int marks=sc.nextInt();  
6         try{  
7             if (marks<0||marks>100){  
8                 throw new IllegalArgumentException();  
9             }  
10            System.out.println("Valid marks:"+marks);  
11        }  
12        catch(IllegalArgumentException e){  
13            System.out.println("Exception:Invalid marks.Marks should  
14        }  
15    }  
16 }
```

Input

120

Output

```
Exception:Invalid marks.Marks  
should be between 0 and 100
```

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

Run

Save

Saving...

Please wait...